

A
Project Report
on
Automobile Buying System

Developed at
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April – 2025

Date: 3rd Apr, 2025**To whomsoever it may concern**

This is to certify that **Ved Thakkar**, a student of **Semester VIII, B.Tech** at **Dharmsinh Desai University, Nadiad**, is currently undergoing an internship at **Maruti Techlabs Private Limited, Ahmedabad** from **December 5, 2024, to April 1, 2025**, as part of his/her academic project work.

The project, titled **"Automobile Buying System,"** is being conducted under the guidance of **Mr. Vishal Savsani** to fulfill academic requirements.

As per the internship offer letter, the internship tenure extends until **June 2025**.

With Best Wishes**For, Maruti Techlabs Pvt. Ltd.,**
Rahul Italiya**RAHUL ITALIYA****(GENERAL HR MANAGER)**

Date: 4th Apr, 2025**To whomsoever it may concern**

This is to certify that **Ayush Patel**, a student of **Semester VIII, B.Tech** at **Dharmsinh Desai University, Nadiad**, is currently undergoing an internship at **Maruti Techlabs Private Limited, Ahmedabad** from **December 5, 2024, to April 1, 2025**, as part of his/her academic project work.

The project, titled **"Automobile Buying System,"** is being conducted under the guidance of **Mr. Vishal Savsani** to fulfill academic requirements.

As per the internship offer letter, the internship tenure extends until **June 2025**.

With Best Wishes**For, Maruti Techlabs Pvt. Ltd.,****RAHUL ITALIYA****(GENERAL HR MANAGER)**

Date: 3rd Apr, 2025**To whomsoever it may concern**

This is to certify that **Yash Vataliya**, a student of **Semester VIII, B.Tech** at **Dharmsinh Desai University, Nadiad**, is currently undergoing an internship at **Maruti Techlabs Private Limited, Ahmedabad** from **December 5, 2024, to April 1, 2025**, as part of his/her academic project work.

The project, titled **"Automobile Buying System,"** is being conducted under the guidance of **Mr. Vishal Savsani** to fulfill academic requirements.

As per the internship offer letter, the internship tenure extends until **June 2025**.

With Best Wishes**For, Maruti Techlabs Pvt. Ltd.,**

Signed by: Vishal Savsani
eSigned using Aadhaar (digio.in)
Date: 2025-04-04 14:03:17 IST

Vishal Savsani**Senior Technical Architect**

CANDIDATE'S DECLARATION

We declare that the final semester report entitled Automobile buying System is our own work conducted under the supervision of the external guide Mr. Vishal Savsani from Maruti Techlabs.

We further declare that to the best of our knowledge the report for B.Tech. Final semester does not contain part of the work which has been submitted for the award of B.Tech. Degree either in this or any other university without proper citation.

Following students were also involved in this project:

Yash Vataliya – Department of CE, DD University

Candidate's Signature:

Candidate's Name : Ved Thakkar

Branch: IT Student ID: 21ITUON159

Candidate's Signature:

Candidate's Name : Ayush Patel

Branch: IT Student ID: 21ITUOS008

DHARMSINH DESAI UNIVERSITY
NADIAD-387001, GUJARAT



CERTIFICATE

This is to certify that the project entitled “Automobile Buying System” is a bonafide report of the work carried out by Mr. Ved Thakkar, Student ID No: 21ITUON159 and Mr. Ayush Patel, Student ID No: 21ITUOS008 of Department of Information Technology, Semester VIII, under the guidance and supervision for the award of the degree of Bachelor of Technology at Dharmsinh Desai University, Nadiad.(Gujarat). They were involved in Project training during the academic year 2024-2025.

Following students were also involved in this project:

Yash Vataliya – CE Department, DD University

Prof. Sunil K. Vithlani

Department of Information Technology,
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Head, Department of Information Technology,
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Date:

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A blend of gratitude, pleasure and great satisfaction is what we feel to convey our indebtedness to all those who have directly and indirectly contributed to the successful completion of the project.

Thanking you.

With Regards,
Ved Thakkar
Ayush Patel

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ABSTRACT

Modern technology makes life easier than it was. While talking about creating any projects for business practices - there is much more work and load than we imagined. Companies or organizations can use different types of communication ways to make development faster, efficient and easy. But for developing a project needs more than just a communication - there may be the case that one need to show their progress, ask views of others, discuss about what to do further, ask upper-level authority about how we will manage future development of the project and enhancements etc.

“Automobile Buying System” offers a fast and hassle-free way to sell used vehicles at absolutely no cost! First buyer has to create campaigns and instant bids. Based on that vehicle is assigned to them. This way Vehicle Trader helps to sell and purchase tens of thousands of vehicles each year, and every day and makes on-the-spot payments to sellers/buyers nationwide. Once the assignment is accepted, the buyer is connected with a carrier who will schedule a convenient time to pick up the vehicle, make on-the-spot payment, and tow the car to the buyer.

COMPANY PROFILE



Maruti Techlabs is a team of software professionals experienced in delivering technology consulting and solutions related to big data, business intelligence, analytics, bot development, mobile platforms, cloud systems, and application development. Maruti Techlabs has a proven track record of exceeding customer expectations through dedication and hard work. We understand the client's business needs and subsequently strive to provide continuous innovation through extensive experience in software development. We combine the cost advantages of an offshore only model with the uncompromising quality standards of a top-notch development team. Our team encourages flexible engagement that quickly aligns with the client's dynamic business models.

It offers robotic process automation (RPA), cloud application development, data analysis, and management services as well as provides WOTNOT products to build chatbot platforms.

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ABBREVIATIONS

API - Application Programming Interface
CSS – Cascading Style Sheet
DB - Database
DOM – Data Object Model
GUI - Graphical User interface
I/O – Input and Output
JS - JavaScript
RAM – Random Access Memory
ROM – Read Only Memory
SEO – Search Engine Optimization
SRS - Software Requirements Specification
UI – User Interface

0. TRAINING TAKEN AT INDUSTRY

0.1 SOFTWARE TRAINING WORK UNDERTAKEN

0.1.1 The Scope and Objective of the software Training

The scope and objective of the software training are listed below:

1. C# training

- C# is both a object-oriented and functional language. The functional part gives it an advantage over the Java programming language allowing you to take advantage of functional constructs and concepts. It is platform independent so we can run java application at any other system.
- Provides many predefine classes and interfaces to wide range.

2. .NET Training

- .NET web application, you can easily use other popular open-source libraries and frameworks as a part of your .NET application. You can build a large web application using ASP.NET Core that can integrate with Angular, TypeScript, And JavaScript
- .NET Provide a consistent, object-oriented programming environment whether object code is stored and executed locally, executed locally but web-distributed, or executed remotely.

3. Basic Frontend Technology Training

- HTML knowledge for creating static web pages.
- CSS for adding different styles to web pages to make it user friendly.
- JavaScript for dynamically add content to web pages and change HTML DOM elements.

0.1.2 Tools and Technology used for software Training

Tools and Technology used for software training are listed below:

1. C# training: Microsoft document, Pluralsight course, Visual studio IDE, and some articles and tutorials for reading.
2. .NET Training: Microsoft document, Pluralsight course, Visual studio IDE, Postman, some articles and tutorials for reading.
3. Basic Frontend Technology Training: HTML, CSS, JavaScript, Udemy, VS Code IDE

0.1.3 Topic Covered in software Training

2- Month Training Period Topics Includes.

Table0.1 Training plan table

Date	Technology/Tools	Weeks
9th Dec to 20th Dec 2024	Basic C# Programming	1-3 Weeks
23th Dec to 3rd Jan 2025	ASP.Net Core	4-6 Weeks
6th Jan to 17th Jan 2025	Web API & MVC	7 Weeks
20th Jan to 31st Jan 2025	RabbitMQ	8-9 Weeks

In Detail of Topics:

Week (1-3)

- Basics of C# Programming.
- Dot Net Core and Framework
- Working with Generic Types
- Different Collections
- Exception Handling
- Asynchronous Programming
- Working with Task Parallel Library

Week (4-6)

- Basics of .Net Core and ASP.Net Core
- Using Azure Key Vault or AWS Parameter Store
- Doing Logging with Third Party Packet Seri log
- Working with middleware and Injecting Dependencies
- Using Built-In Middleware like CORS, Authentication, Static Files, Routing
- Registering and using Services
- Working with EF Core

Week (7)

- Working with Web APIs
- Testing APIs on Postman
- Setting up MVC Application
- Adding Background Tasks
- Building Worker Services and running them in production
- Working on SQL

Week (8-9)

- Introduction to RabbitMQ
- Understanding Exchanges and AMQP Messaging Protocols
- Implementing Microservices and Message Queuing
- Mass Transit in RabbitMQ
- Topic Bases Publisher and Subscriber

During Training, they have assigned a mentor for our queries. We had a daily scrum meeting with our mentor. In scrum, the meeting mentor asks questions from the last day's work and sometimes he also asks for the demo of work. They conduct three tests. Definition for the tests is given below.

0.2 INDUSTRIAL TRAINING WORK UNDERTAKEN**Test 1:****Title: Data insertion**

Write a console application to generate a strong password and store it in the database. We want to have such 30 million passwords in the database table.

- Password's length is 20 chars
- The password contains a combination of uppercase, lowercase, numeric and special chars
- The database doesn't contain duplicate entries
- In the single transaction, bulk entries are not allowed i.e you can insert/update/get a single record from/to the database
- You can use SQL Server or MY SQL.
- Completing the entire job in minimum time would be preferred.

Once the job is complete, print time is taken in the console.

Github Code : <https://github.com/VedThakkar/PasswordGenerator.git>

Test 2:

Title: Data Migration Utility

Create a Data Migration Utility that reads data from the Source table, manipulates records, and stores them in the Destination table.

- Let's have two tables given below
 - Source Table
 - Columns ID (primary key), First Number, Second Number
 - Destination Table
 - ID, Foreign Key (from Source Table), Sum
- Add 1 million records in Source Table
- Create a console application to migrate records from Source Table to Destination Table after summing up First Number and Second Number
- Create a function in C# that takes First Number and Second Number as input and sums up. It has a delay of 50 milliseconds to mimic as its I /o (third-party HTTP API) operation
- The app takes range (start number and end number) as input for which it migrates related records from the ID column. Let's call it Batch.
- At any point in time, you should be able to cancel migration by just typing the
 - "CANCEL" keyword in the console.
- One can see the status of migration by typing the "STATUS" keyword in the console.
 - It shows all completed, cancelled, and ongoing migrations.
- You cannot insert more than 100 records in a database in a single API call.
- You can use SQL Server for data storage.
- You may have to use collections to store migration status.

Github Code : <https://github.com/VedThakkar/UtilityMigrationProject.git>

Test 3:

Title: .NET REST API Application(microservice)

Let's build a REST API using .NET 8.

- Use Entity Framework Core to insert/update/delete records from the database.

- Use SQL Server.
- The application should log the request and response of the API in the console.
- Log the time taken by API using middleware.
- Dependencies need to be injected throughout the app.
- Use Swagger to document your APIs
- Connection string/secret need to be made configurable
- Demonstrate GET, POST, PUT, PATCH, and DELETE APIs

Github Code : <https://github.com/VedThakkar/CityInfoWebAPI.git>

1. INTRODUCTION

1.1 PROJECT DETAILS

Automobile Buying System offers a fast and hassle-free way to sell used vehicles at absolutely no cost! First buyer has to create campaigns and instant bids. Based on that vehicle is assigned to them. This way Vehicle Trader helps to sell and purchase tens of thousands of vehicles each year, and every day and makes on-the-spot payments to sellers/buyers nationwide. Once the assignment is accepted, the buyer is connected with a carrier who will schedule a convenient time to pick up the vehicle, make on-the-spot payment, and tow the car to the buyer.

1.2 PURPOSE

Automobile Buying System is a next generation ecommerce platform for pre-owned autos like cars and bikes. It provides the best experience for car buyers by offering a wide assortment of certifies cars that are home delivered in a click of a button while sellers get the best price of their vehicle's instant

Automobile Buying System is single stop for buying or selling used cars, all over India. It brought together cutting-edge technology with country-wide partners and more importantly, deep understanding of what buyers and sellers need. It solves all pain points associated with selling an existing car, or purchasing a pre-loved one. Whether customer buying or selling, Customer will get a quick, easy, fair, transparent, hassle (and haggle) free process.

1.3 SCOPE

Automobile Buying System is one type of E-Commerce Site for Used Cars selling and buying. The current system can be extended to allow the users to create accounts and create campaigns and instant bids. They can save search vehicles based on different categories. The users could subscribe for price alerts which would enable them to receive messages when price for products fall below a particular level. The current system is confined only to the shopping cart process. It can be extended to have an easy to use check out process. Users can have multiple shipping and billing information saved. During checkout they can use the drag and drop feature to select shipping and billing information.

1.4 OBJECTIVE

The objective of this project is to give assignment to buyer based on the vehicle data, campaigns and instant bids entered by the buyer and past auction data and if buyer accept that assignment then further process will be done.

1.5 TECHNOLOGY AND LITERATURE REVIEW

1.5.1 .NET Backend Technology

The .NET Framework is a software development platform that was introduced by Microsoft in the late 1990 under the NGWS. On 13 February 2002, Microsoft launched the first version of the .NET Framework, referred to as the .NET Framework 1.0.

It is a virtual machine that provide a common platform to run an application that was built using the different language such as C#, VB.NET, Visual Basic, etc. It is also used to create a form based, console-based, mobile and web-based application or services that are available in Microsoft environment. Furthermore, the .NET framework is a pure object oriented, that similar to the Java language. But it is not a platform independent as the Java. So, its application runs only to the windows platform. The main objective of this framework is to develop an application that can run on the windows platform.

1.5.2 SQL

SQL stands for Structured Query Language. SQL is used to create, remove, alter the database and database objects in a database management system and to store, retrieve, update the data in a database. SQL is a standard language for creating, accessing, manipulating database management system. SQL works for all modern relational database management systems, like SQL Server, Oracle, MySQL, etc.

1.5.3 AngularJS - Frontend Technology

AngularJs is a JavaScript open-source front-end framework that is mainly used to develop single-page web applications (SPAs). It is a continuously growing and expanding framework which provides better ways for developing web applications. It changes the static HTML to dynamic HTML. Its features like dynamic binding and dependency injection eliminate the need for code that we have to write otherwise. AngularJs is rapidly growing and because of this reason, we have different versions of AngularJs with the latest stable being 1.7.7. It is also important to note that Angular is different from AngularJs. It is an open-source project which can be freely used and changed by anyone. It extends HTML attributes with Directives, and data is bound with HTML.

1.5.4 HTML / CSS / JavaScript

HTML stands for Hyper Text Mark-up Language. HTML describes the structure of Web pages using mark-up. HTML elements are the building blocks of HTML pages. HTML elements are represented by tags.

CSS stands for Cascading Style Sheets. CSS describes how HTML elements are to be displayed on screen, paper, or in other media. CSS saves a lot of work. It can control

the layout of multiple web pages all at once. External style sheets are stored in CSS files.

JavaScript is a dynamic computer programming language. It is lightweight and most commonly used as a part of web pages, whose implementations allow client-side script to interact with the user and make dynamic pages. It is an interpreted programming language with object-oriented capabilities.

2. PROJECT MANAGEMENT

2.1 FEASIBILITY STUDY

2.1.1 Technical Feasibility

The system is technically feasible. We have the necessary knowledge and technology, as we've confirmed that the Buyer Site can provide assignments based on Valuation. Coordination between development and testing has been maintained throughout.

2.1.2 Time Schedule Feasibility

We checked whether our system can be ready in time without any error. We have planned the project in 4 Sprints, the actual development is supposed to be completed within the first 3 sprints, the last sprint was reserved for testing and bug fixes after testing and to implement additional features in case client requirement alters.

2.1.3 Operational Feasibility

How the project will work and who will use it, all such concerns arise in this phase. We have to study the existing system's problem and is it worth solving or not.

2.1.4 Implementation Feasibility

We identified that Buyer Site is using REST-Endpoints to handle all its web Interfaces the Valuation platform also provides REST endpoints to provide the data obtained on the browser, and this is beneficial to Give Instant assignment corresponding to Details of Vehicle. Hence implementation feasibility is achieved.

2.2 PROJECT PLANNING

2.2.1 Project Development Approach and Justification

Agile Model



Fig. 2.1 Agile Model

Ref. <https://www.researchgate.net/figure>

1. **Requirements gathering:** In this phase, you must define the requirements. You should explain business opportunities and plan the time and effort needed to build the project. Based on this information, you can evaluate technical and economic feasibility.
2. **Design the requirements:** When you have identified the project, work with stakeholders to define requirements. You can use the user flow diagram or the high-level UML diagram to show the work of new features and show how it will apply to your existing system.
3. **Construction/ iteration:** When the team defines the requirements, the work begins. Designers and developers start working on their project, which aims to deploy a working product. The product will undergo various stages of improvement, so it includes simple, minimal functionality.
4. **Testing:** In this phase, the Quality Assurance team examines the product's performance and looks for the bug.
5. **Deployment:** In this phase, the team issues a product for the user's work environment.
6. **Feedback:** After releasing the product, the last step is feedback. In this, the team receives feedback about the product and works through the feedback.

2.2.2 Project Plan

In managing any project, the whole plan of the project is made before its actual implementation. The plan of the project helps the team to work as per the schedule and helps to successfully complete the project. To plan a project the main requirements that are calculated are cost, duration, effort, scheduling, manpower, resource allocation, risk management etc. We plan on taking a time-based approach to deliver this product on time. To begin with, we create a set of deadlines to complete certain tasks. This will give us a general idea on how to meet the final deadline on time.

2.2.3 Milestones and Deliverables

Milestones:

- Milestone is an end-point of the software process activity.
- At each milestone there should be formal output, such as a report, that can be represented to the management. The weekly report is submitted to the project guide, which includes a day-to-day work report.
- Milestone represents the end of the distinct, logical stage in the project.

Deliverables:

- Deliverable-s is a project report that is delivered to the administrator of the project.

2.2.4 Roles & Responsibilities

Table 2.1 Roles and Responsibilities

Name	Role				
	Analysis	Designing	Coding	Testing	Documentation
Ved Thakkar	✓	✓	✓	✓	✓
Ayush Patel	✓	✓	✓	✓	✓
Yash Vataliya	✓	✓	✓	✓	✓

3. SYSTEM REQUIREMENTS STUDY

3.1 STUDY OF CURRENT SYSTEM

In the existing system, the registered users and unregistered users can buy/sell use cars with some searching criteria like colour, company. In this system, if the visitors can visit the website once then message will not be going in their mobile/email by the existing system. The existing system is paid and, in this system, commission is taken for buying/selling used cars.

3.2 PROBLEM AND WEAKNESS OF CURRENT SYSTEM

In our new system, buyer can search used cars with many criteria like company, budget, colour, model, city, year etc. In our system, we want to add new module campaigns and instant bids. In our new system searching criteria will be strong with year, budget etc. The new feature of our system will message. In this module, Visitor can visit in our website once, when the new used car will deal in the system, message/mail will be passed in their messages.

3.3 USER CHARACTERISTICS

There is one user in our module that is buyer who wants to buy car. Buyer can visit website and enter details like year, branch, Model, Transmission type, variant etc. After that he can add campaigns and instant bids and user get their assignment of car based on that condition.

3.4 HARDWARE AND SOFTWARE REQUIREMENTS

3.4.1 Hardware Requirements

- Intel x86-bit chip architecture, 12 CPU cores at 2GHz Processor
- 4GB ram
- 150 Mb of available hard disk

Storage and processing power of these resources are dependent on the scaling of the application. The software can run on minimal hardware.

3.4.2 Software Requirements

- Frontend : Web Browser
- Backend : SSMS
- Server : IIS Server
- IDE : JetBrains Rider

3.5 Constraints

3.5.1 REGULATORY POLICIES

Some regulatory policies that should be followed by the developers include – abiding to certain coding standards, following best practices, logging various milestones and crashes, commenting in the code appropriately, using a version control system with its best practices.

3.5.2 HARDWARE LIMITATIONS

There are no specific hardware limitations, the software is least likely to run into hardware issues.

3.5.3 INTERFACES TO OTHER APPLICATIONS

This web portal provides the opportunity to clients to see their project progress and for employees to store data at one place and also see their progress, so there is no need for the interface with the other applications.

3.5.4 PARALLEL OPERATIONS

Numerous parallel operations take place in the working of the Web portal. When the client website is executed, several AJAX calls are made asynchronously, which are used for the collection of data like tracking the location of the user if permitted.

3.5.5 RELIABILITY REQUIREMENTS

In order for the application to be reliable and accurate, hardware uptime is extremely important. If at any time the server is down, it may fail to collect the user data for the downtime. This may result in inaccurate data in the database. This will make the software unreliable and take away the entire purpose of the application.

3.5.6 CRITICALITY OF THE CONSIDERATION

As per the known issues, there does not exist any problems with the system that can make the system so critical that it can affect the working software or the underlying hardware.

3.5.7 SAFETY AND SECURITY CONSIDERATION

The data collected from the client, may contain some sensitive information regarding the project. If this data is stolen, it may put the company in jeopardy. Thus, the data stored should undergo an adequate amount of encryption and security measures

3.6 ASSUMPTION AND DEPENDENCIES

Some known assumptions and dependencies are mentioned below:

- The web server and the database server work with perfect uptime.

4. SYSTEM ANALYSIS

4.1 REQUIREMENT OF NEW SYSTEM

4.1.1 Use-Case Diagram

A Use case Diagram is a dynamic or behaviour diagram in UML. Use Case Diagrams model the functionality of a system using actors and use cases. Use cases are a set of actions, services, and functions that the system needs to perform. In this context, a “System” is something being developed or operated, such as a website. The “Actors” are people or entities operating under defined roles within the system.

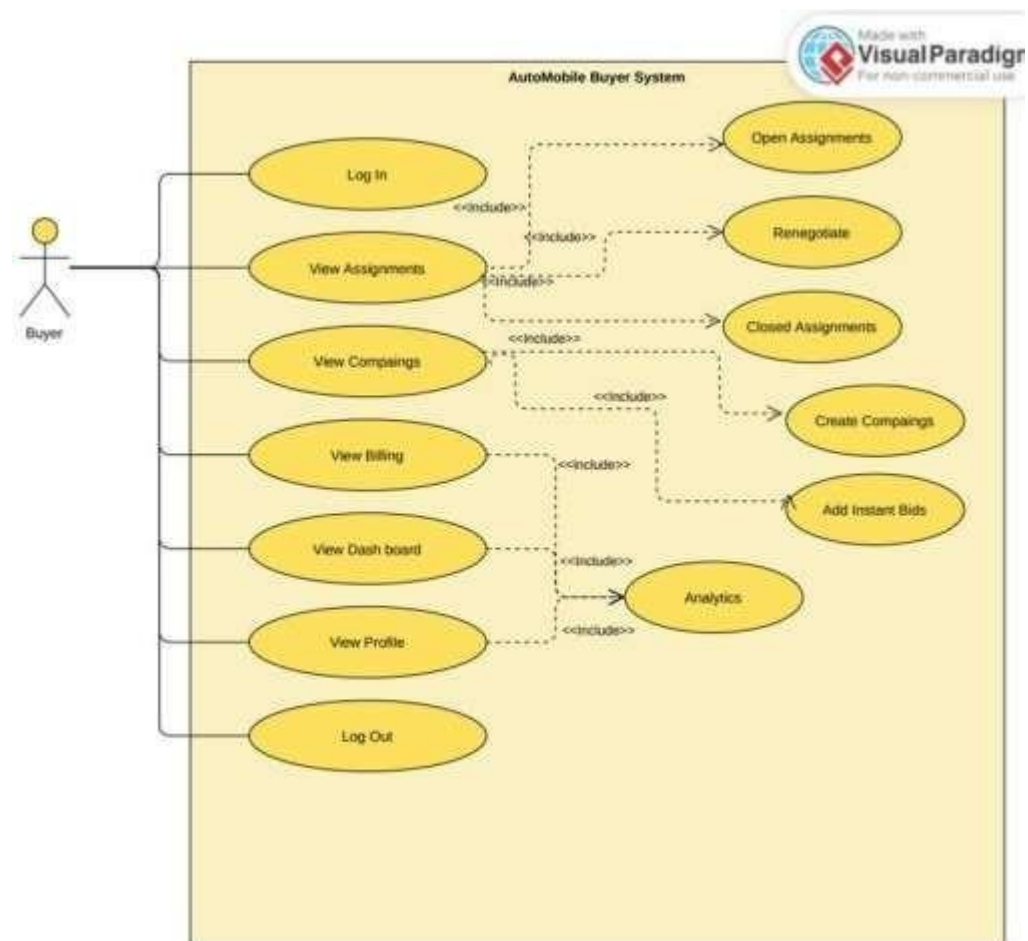


Fig. 4.1 Use Case Diagram

4.1.2 User Requirements

For using this application there is no need for technical knowledge, thus any user with their needs can use this application. User requirements include minor details, but most importantly users must be aware that the system works properly with full availability, reliability, security, and safety. The user’s responsibility is as follows: The user should know how to use the application and should adhere to the guidelines and prescribed standards.

4.1.3 System Requirements

Functional Requirements:

R1: Manage Instant Bid

Description: The user needs to create an instant bid so that the user can get an assignment based on the created instant bid.

R1.1: Add Instant Bids

Description: The user needs to enter the following details and the system will Add Instant Bid.

Input:

- Name
- Campaign List
- Year
- Make
- Model
- Title
- Vehicle Condition
- Mileage

Output: Instant Bid Added Successfully.

R1.2: Change Instant Bid Status

Description: user can change the Instant Bid status to Active, Pause, and Delete.

Input: select status from dropdown List and Click update

Output: status changed.

R1.3: View Instant Bid

Description: user can select an instant bid from the list and view instant bid details.

Input: Instant Bid Id.

Output: Instant bid details.

R2: Manage Buyer Account

Description: Buyer account contains details of Buyer. we can fetch, create, and update Buyer details.

R2.1: Get Buyer Details

Description: fetch Buyer details on BuyerId

Input: Buyer Id.

Output: Buyer Details like first name, last name, email, phone no.

R2.2: Update Buyer Details

Description: Buyer can update details on the profile page.

Input: Buyer Id, buyer object.

Output: Updated Buyer details.

R2.3: Change Password

Description: Buyer can change their password by clicking the change password button.

Input: new password, confirm the new password.

Output: password changed.

R2.5: Create New Buyer

Description: if Buyer wants to create their account on the system

Input: Buyer account details.

Output: Buyer account created successfully.

R3: Manage Campaign

Description: A campaign is a region where a buyer gets an assignment based on an instant bid.

R3.1: Create Campaign

Description: Create a Campaign for the selection of a particular region.

Input: Campaign details.

Output: Campaign Created successfully.

R3.2: Get All Campaign

Description: Buyer User can see all campaigns.

Input: Click on Campaign Tab.

Output: Display List of all Campaign.

R3.3: Get Campaign Detail

Description: Buyer User can see particular campaign details.

Input: Click on Campaign.

Output: Campaign page which contains campaign Details.

R4: Registration

Description: The new user needs to get themselves registered when the user opens the Site for the first time.

R4.1 Taking data from the user:

Input: The user needs to provide the following information while registering.

Name

Mobile Number

Email ID

Password(for the Website) Preferred location.

Processing State: The data will be saved in the database when the user clicks on the next button.

Output: The user data is saved and he is directed to the new page where he needs to enter the OTP sent on his registered mobile number.

R4.2 OTP Verification:

Input: The user needs to enter the OTP sent on the registered mobile number.

Processing State: The OTP will be checked for its correctness and validity.

Output: Registration is successful and the user is directed to the Dashboard page.

R5: Login

Prerequisite:

For the users who are already registered.

R5.1: Successful Login

Description: The user can log in to his account with his registered mobile number/email id and password.

Input: On the login page, the user needs to enter his email id or registered mobile number and password which is set by him.

Processing State: The email id / mobile number and password will be checked with the database of the application.

Output: Successful Login. Users will be directed to the homepage of the application.

R5.2: Forgot Password

Description: In case the user forgets his password then he can use this feature.

Input 1: Click on the Forgot Password option on the login page.

Output 1: The user is directed to the page where he needs to enter his registered mobile number.

Input 2: Enter the registered mobile number in the text field and click on the next button.

Processing State: The user's password is fetched from the database server.

Output 2: The user gets his password on his registered mobile number which he can use for the login.

R6: Manage Open Assignments

Description: In this user can view assignments that are already created but are not completed or cancelled

Input 1: Click on the Open Assignments option on the Home Screen.

Output 1: The user is directed to the page where he is displayed a list of Open assignments.

R6.1: Reschedule Open assignment

Description: The user can reschedule selected open assignments

Input: Select Open Assignment and Click on Reschedule option user needs to enter new pickup date

Processing State: The new pickup date will be modified in the database of the application.

Output: Successfully Rescheduled date & Users will be Displayed modified List of the assignments.

R6.2: Cancel Open assignment

Description: The user can cancel selected open assignments

Input: Select Open Assignment and Click on Cancel option user needs to enter Cancellation Issue & Comment

Processing State: The Assignment will be removed from the database of the application.

Output: Successfully Cancelled Assignment & Users will be Displayed modified List of the assignments.

R7: Manage Closed Assignments

Description: In this user can view assignments that have been completed or cancelled

Input 1: Click on the Closed Assignments option on the Home Screen.

Output 1: The user is directed to the page where he is displayed a list of Closed assignments.

R7.1: Print Completed assignment

Description: The user can print Details of Completed assignment

Input: Select Completed Assignment and Click on Print option

Processing State: Print of Completed Assignments with all its details will be generated from Data Fetched by database

Output: Print will be generated in PDF Format. And will be displayed to the user.

Non-Functional Requirements:

Non-functional requirements describe how a system must behave and establish constraints of its functionality. This type of requirement is also known as the system's quality attributes.

- **Usability:**

Usability defines how difficult it will be for a user to learn and operate the system.

- **The efficiency of use:**

The average time it takes to accomplish a user's goals, how many tasks a user can complete without any help, the number of migrations completed without errors, etc.

- **Intuitiveness:**

How simple it is to understand the interface, buttons, headings, etc.

- **Low perceived workload:**

How many attempts are needed by users to accomplish a particular task.

- **Security:**

Security requirements ensure that the software is protected from unauthorized access to the system and its stored data. It considers different levels of authorization and authentication across different user roles. For instance, data privacy is a security characteristic that describes who can create, see, copy, change, or delete information. Security also includes protection against viruses and malware attacks.

- **Reliability:**

Reliability defines how likely it is for the software to work without failure for a given period. Reliability decreases because of bugs in the code, hardware failures, or problems with other system components. To measure software reliability, you can count the percentage of operations that are completed correctly or track the average period the system runs before failing. Example: The database update process must roll back all related updates when any update fails.

- **Performance:**

Performance is a quality attribute that describes the responsiveness of the system to various user interactions with it. Poor performance leads to a negative user experience. It also jeopardizes system safety when it's overloaded.

- **Availability:**

Availability is gauged by the period of time that the system's functionality and services are available for use with all operations. So, scheduled maintenance periods directly influence this parameter. And it's important to define how the impact of maintenance can be minimized. When writing the availability requirements, the team has to define the most critical components of the system that must be available at all times. You should also prepare user notifications in case the system or one of its parts becomes unavailable. Example: New module deployment mustn't impact the front page, visualization/alert pages, and migration mustn't take a long time.

- **Scalability:**

Scalability requirements describe how the system must grow without a negative

System Analysis influence on its performance. This means migrating more assets, processing more data. Scalability has both hardware and software implications. For instance, you can increase scalability by adding memory, servers, or disk space.

4.2 FEATURE OF NEW SYSTEM

In our new system, buyer can search used cars with many criteria like company, budget, colour, model, city, year etc. In our system, we want to add new module as a comparison car with company and budget. In our new system searching criteria will be strong with year, budget etc. The new feature of our system will message. In this module, Visitor can visit in our website once, when the new used car will deal in the system, message/mail will be passed in their messages.

4.3 SYSTEM ACTIVITY (ACTIVITY DIAGRAM)

An activity diagram is a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system. An activity diagram is essentially an advanced version of the flowchart that models the flow from one activity to another activity

Login Activity Diagram

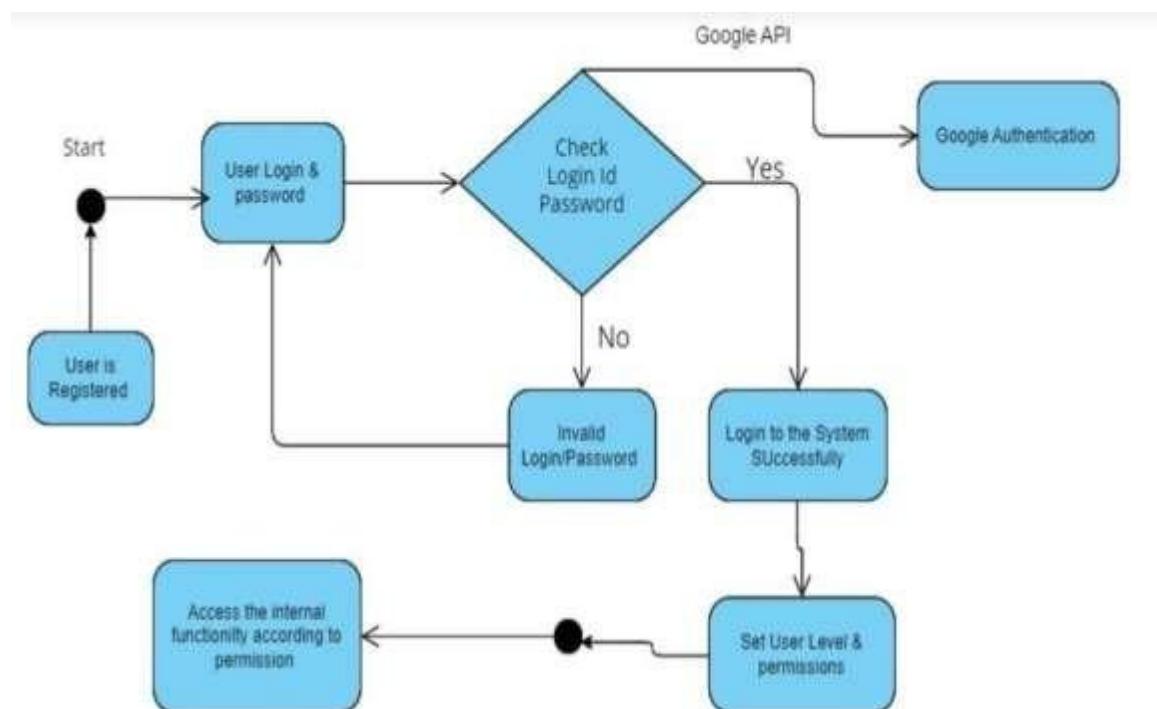


Fig 4.2Login Activity Diagram

Open Assignments Diagram

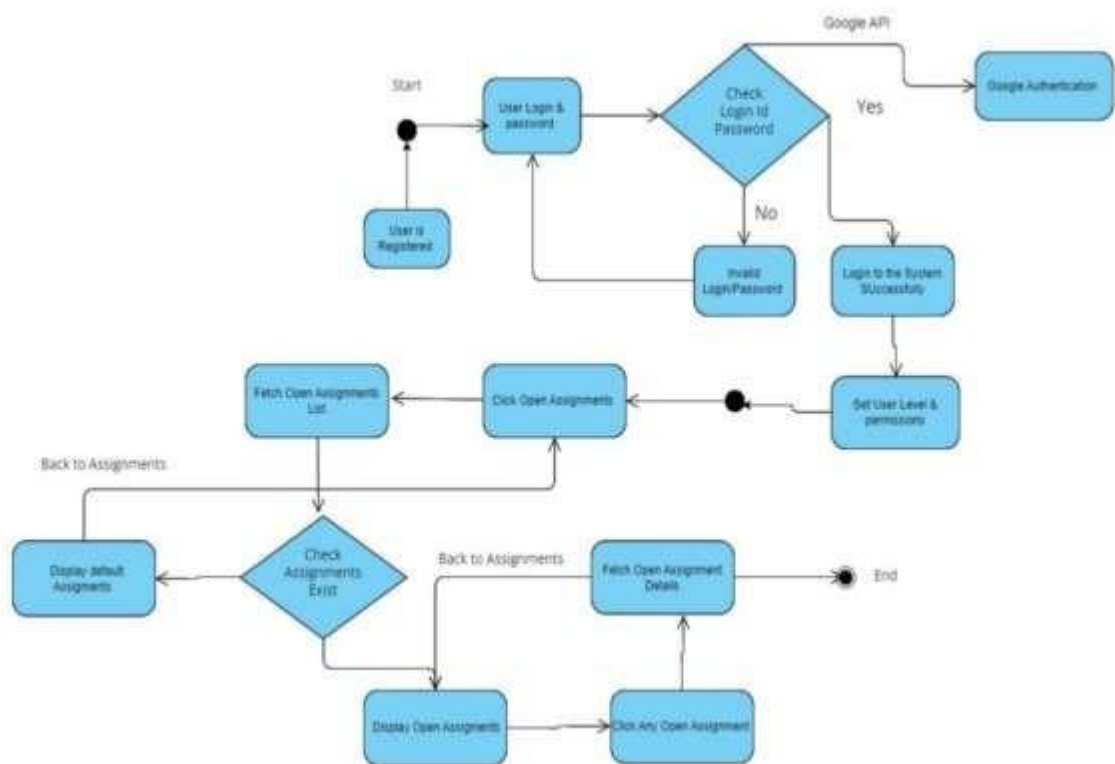


Fig 4.3 Open Assignment Activity Diagram

Add Campaign Activity

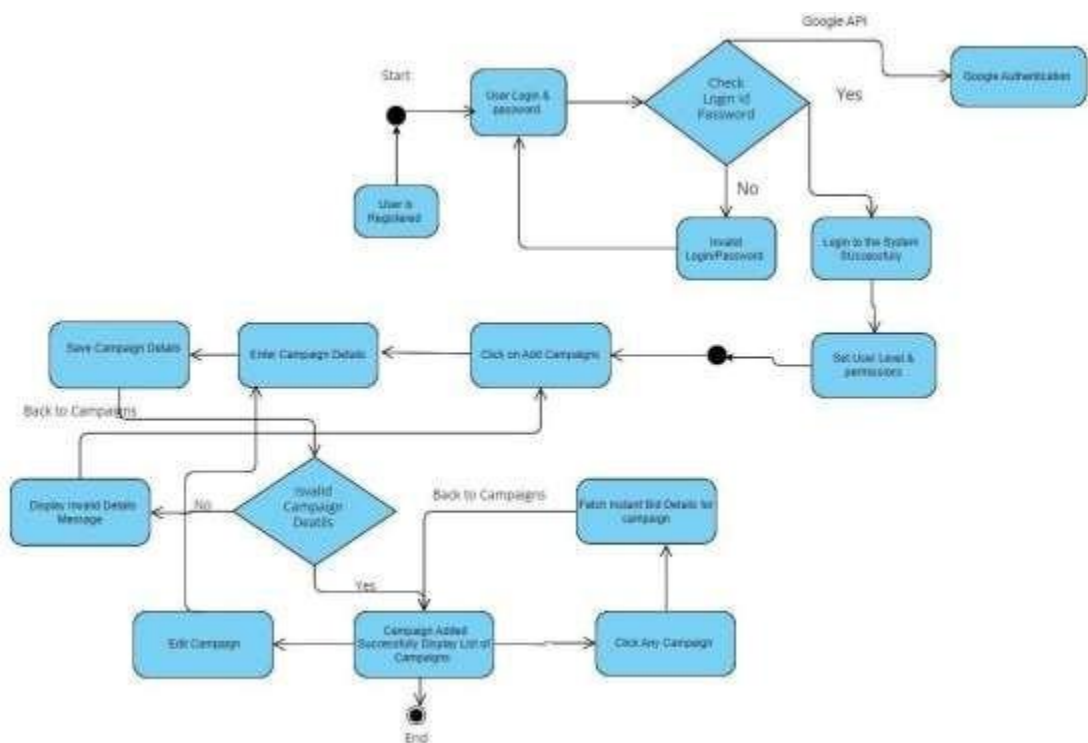


Fig 4.4 Add Campaign Activity Diagram

4.4 DATA MODELING – ER DIAGRAM

An ER diagram shows the relationship among entity sets. An entity set is a group of similar entities and these entities can have attributes. An entity is a table or attribute of a table in a database, so by showing relationships among tables and their attributes, the ER diagram shows the complete logical structure of a database.

An ER diagram has three main components:

1. Entity
2. Attribute
3. Relationship

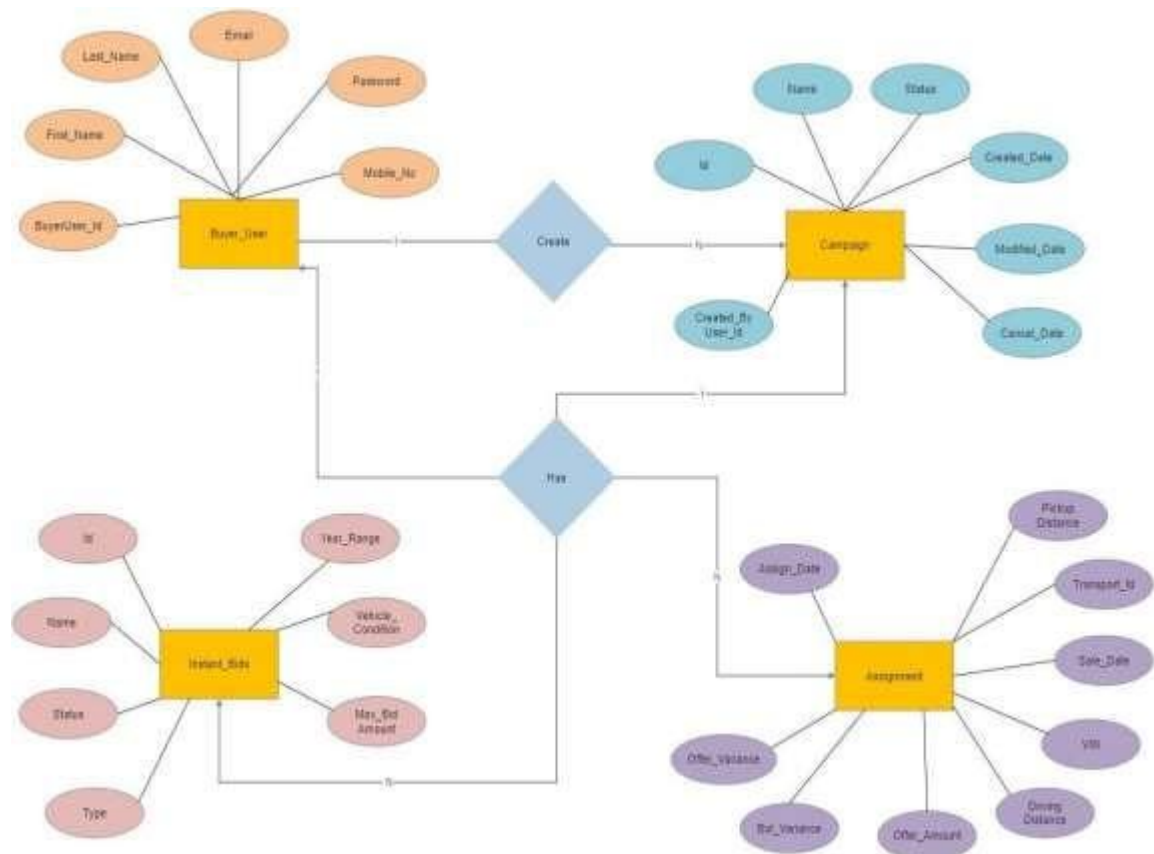


Fig 4.4 ER Diagram

5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE DESIGN

Systems design is the process of defining the architecture, modules, interfaces, and data for a system to satisfy specified requirements. Systems design could be seen as the application of systems theory to product development. There is some overlap with the disciplines of systems analysis, systems architecture and systems engineering.

Architectural Design

The architectural design of a system emphasizes the design of the systems architecture that describes the structure, behaviour and more views of that system and analysis.

Logical Design

The logical design of a system pertains to an abstract representation of the data flows, inputs and outputs of the system. This is often conducted via modelling, using an over-abstract (and sometimes graphical) model of the actual system. In the context of systems, designs are included.

Physical Design

The physical design relates to the actual input and output processes of the system. This is explained in terms of how data is input into a system, how it is verified/authenticated, how it is processed, and how it is displayed. In physical design, the following requirements about the system are decided.

- Input requirement
- Output requirements
- Storage requirements
- Processing requirements

User Interface Design is concerned with how users add information to the system and with how the system presents the information back to them. Data Design is concerned with how the data is represented and stored within the system. Finally, Process Design is concerned with how data moves through the system, and with how and where it is validated, secured and/or transformed as it flows into, through and out of the system. At the end of the system design phase, documentation describing the three subtasks is produced and made available for use in the next phase.

Physical design, in this context, does not refer to the tangible physical design of an information system. To use an analogy, a personal computer's physical design involves input via a keyboard, processing within the CPU, and output via a monitor, printer, etc. It would not concern the actual layout of the tangible hardware, which for a PC would be a monitor, CPU, motherboard, hard drive, modems, video/graphics cards, USB slots, etc. It involves a detailed design of a user and a product database structure processor and a control processor.

5.1.1 Class Diagram

A class diagram is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects

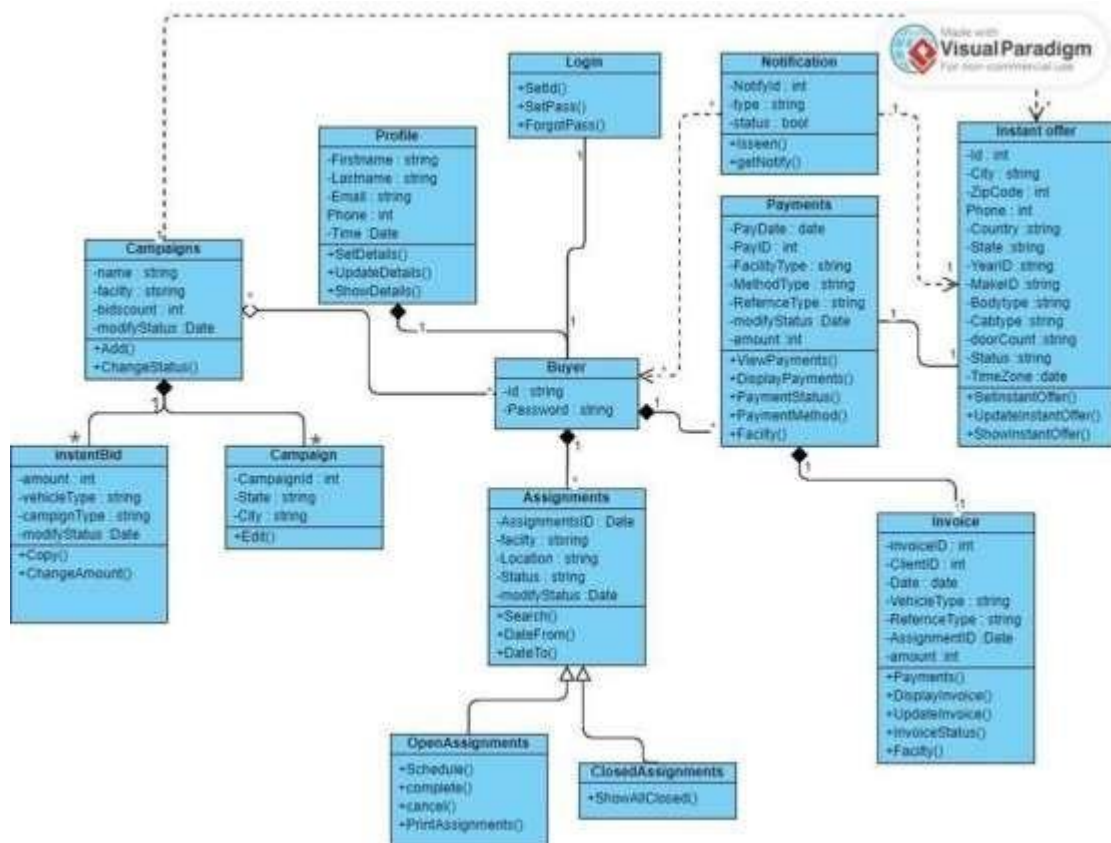


Fig 5.1 Class Diagram

5.1.2 Sequence Diagram

Sequence Diagrams are interaction diagrams that detail how operations are carried out. They capture the interaction between objects in the context of a collaboration. Sequence Diagrams are time focus and they show the order of the interaction visually by using the vertical axis of the diagram to represent time what messages are sent and when

Add campaignDiagram

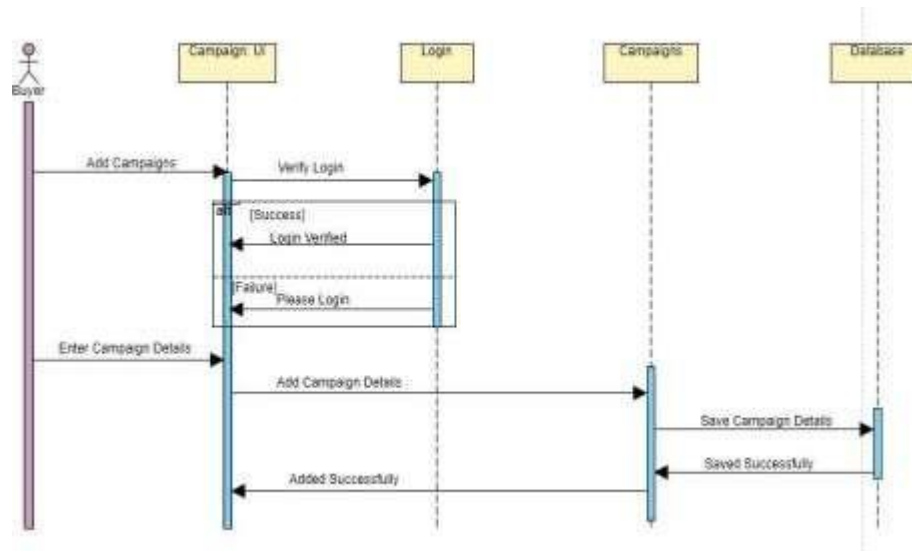


Fig 5.2Add Campaign Diagram

Open Assignment Diagram

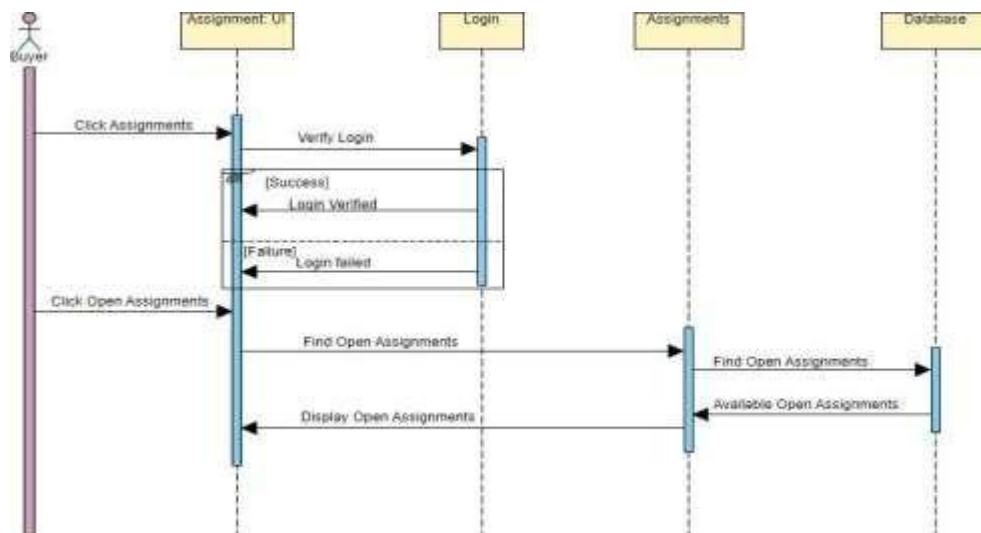


Fig 5.3Open Assignment Diagram

Schedule Assignment Diagram

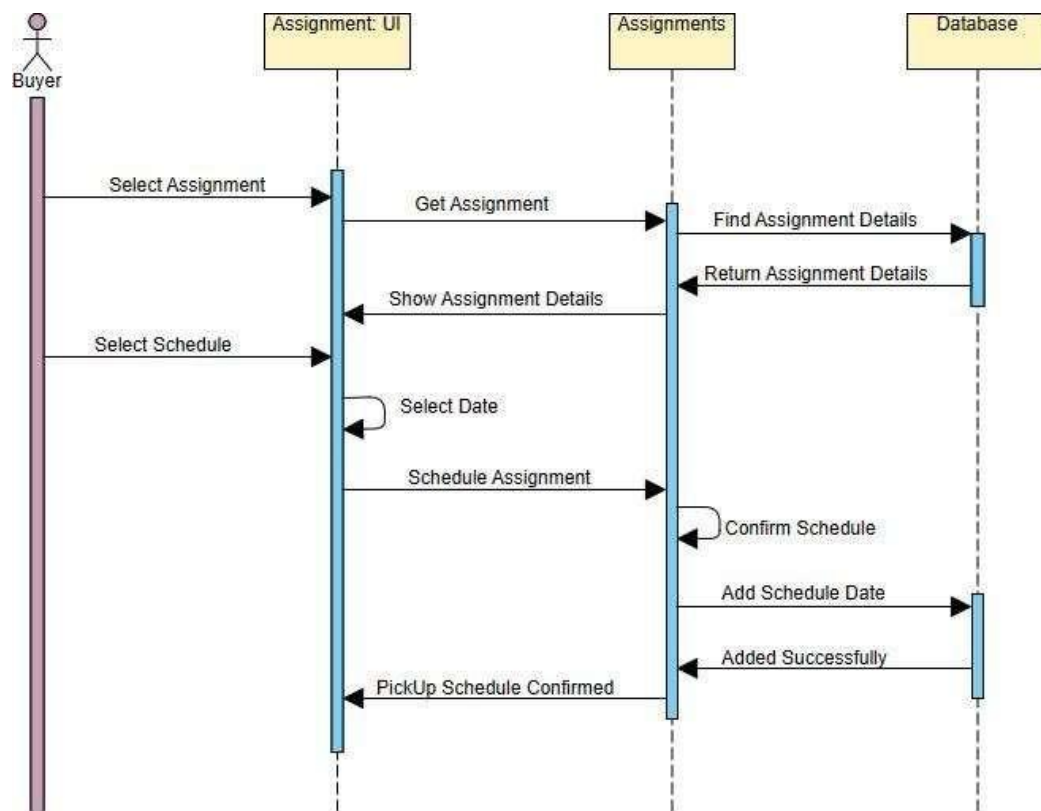


Fig 5.4 Schedule Assignment Diagram

5.2 DATABASE DESIGN

5.2.1 Table and Relationship (Detail ER Diagram)

An ER diagram shows the relationship among entity sets. An entity set is a group of similar entities and these entities can have attributes. An entity is a table or attribute of a table in a database, so by showing relationships among tables and their attributes, the ER diagram shows the complete logical structure of a database.

An ER diagram has three main components:

1. Entity
2. Attribute
3. Relationship

5.2.2 Data Dictionary

Here we had used SQL Server for database

Table: 5.1 Buyer User table

BuyerUser

	Column Name	Data Type	Allow Nulls
	BuyerUserId	bigint	☐
	BuyerId	int	☐
	FirstName	varchar(40)	☐
	LastName	varchar(80)	☐
	UserStatusTypeId	smallint	☐
	UserAccessTypeId	smallint	☐
	Email	varchar(80)	☐
	Password	nvarchar(1000)	☒
	PasswordToken	varchar(32)	☒
	IsPasswordChangeRequi...	bit	☐
	SalesforceId	varchar(18)	☒
	Mobile	varchar(10)	☒
	PhoneExt	varchar(5)	☒
	Phone	varchar(10)	☒
	IsTextUpdateEnabled	bit	☐
	TimeZone	varchar(3)	☐
	CreatedDateTime	datetime	☐
	LastModifiedDateTime	datetime	☐
	DeactivatedDate	date	☒
	CreatedByUserId	int	☐
	CreatedByUserId	bigint	☐
	LastModifiedByUserId	int	☐
	LastModifiedByUserId	bigint	☐
	FullName		☐

Table 5.2 Campaign Table

Campaign

	Column Name	Data Type	Allow Nulls
	Id	int	☐
	Name	varchar(250)	☐
	AcceptWithoutTitle	bit	☐
	Status	smallint	☐
	BuyerId	int	☐
	OLD_CreatedByUserId	int	☒
	CreatedDate	datetime	☐
	OLD_LastModifiedByUse...	int	☒
	LastModifiedByContactId	int	☒
	LastModifiedDate	datetime	☒
	DeletedDate	datetime	☒
	CreatedByUserTypeId	int	☒
	CreatedByUserId	bigint	☒
	LastModifiedByUserTypeId	int	☒
	LastModifiedByUserId	bigint	☒
			☐

Table 5.3 Instant Bids Table

InstantBids

	Column Name	Data Type	Allow Nulls
	Id	int	☐
	Name	varchar(250)	☐
	Status	smallint	☐
	Type	int	☐
	BidAmount	money	☐
	YearRange	int	☐
	FromYear	int	☐
	ToYear	int	☒
	FromDisplayYear	int	☐
	ToDisplayYear	int	☒
	AnyVehicleCondition	bit	☐
	ExcludeMissingEngineOr...	bit	☐
	ExcludeAllOrSomeTiresR...	bit	☐
	VehicleRunAndDrive	bit	☒
	LittleOrNoExtDamage	bit	☒
	Mileage	int	☐
	AllMakeModel	bit	☐
	AllBodyStyle	bit	☐
	CampaignId	int	☐
	OLD_CreatedByUserId	int	☒
	CreatedDate	datetime	☐
	OLD_LastModifiedByUse...	int	☒
	LastModifiedByContactId	int	☒
	LastModifiedDate	datetime	☒
	DeletedDate	datetime	☒
	MaxBidAmount	money	☒
	CreatedByUserTypeId	int	☒
	CreatedByUserId	bigint	☒
	LastModifiedByUserTypeId	int	☒
	LastModifiedByUserId	bigint	☒
			☐

Table 5.4 Byuers Bid Table

Tbl_BuyersBid

	Column Name	Data Type	Allow Nulls
► 🔑	Id	bigint	☐
	BidID	nvarchar(25)	☒
	BidAmount	smallmoney	☐
	CreatedDate	datetime	☐
	CacheTime	smallint	☐
	BuyerBidKey	varchar(500)	☒
	BuyerID	int	☒
			☐

Table 5.5 Buyer to Seller Table

Tbl_BuyerToSellerEmails

	Column Name	Data Type	Allow Nulls
► 🔑	Id	bigint	☐
	AssignmentId	int	☐
	ContactId	bigint	☐
	SentDate	datetime	☐
	Message	text	☐
	FromEmail	varchar(100)	☐
			☐

AssignmentLimit	int	☒
AssignmentLimitCap	int	☒
MinimumInvoiceDueDate	date	☒
NoTitleRequirement	varchar(250)	☒
ParentAccountID	int	☒
Industry	varchar(100)	☒
LastAssignmentDate	datetime	☒
IsLimited	bit	☐
AccountRecordType	varchar(50)	☐
IsLimitAPIBidRequestsTo...	bit	☐
IsLimitAPIBidRequeststol...	bit	☐
DealerLicenseVerifiedDate	datetime	☒
RenegotiationEnabled	bit	☐
VinRequirement	int	☐
FeeTierType	int	☐
ReadRequirement	bit	☐
Website	varchar(255)	☒
AssignmentNotification	varchar(25)	☒
CloseWonBy	bigint	☒
LastModifiedDateTime	datetime	☐
InactivityWarningEmailS...	datetime	☒
PaymentTermId	smallint	☐
LastReviewDate	date	☒
LastTouchpointDate	date	☒
Type	varchar(50)	☐
DropOffInstruction	varchar(500)	☒
🔑 AFF_ID	int	☐
IsMessageTemplateEnabl...	bit	☐
GeocodeLocationType	varchar(30)	☒

Table 5.6 Buyer Master Table

Tbl_Buyers

	Column Name	Data Type	Allow Nulls
▶	AFF_COMPANYNAME	varchar(255)	☒
	AFF_ADDRESS1	varchar(100)	☒
	AFF_CITY	varchar(50)	☒
	AFF_ZIPCODE	varchar(50)	☒
	AFF_StateCode	char(2)	☒
	AFF_Country	varchar(50)	☒
	AFF_MailingAddress1	varchar(100)	☒
	AFF_MailingCity	varchar(50)	☒
	AFF_MailingZipcode	varchar(50)	☒
	AFF_MailingStateCode	char(2)	☒
	AFF_MailingCountry	varchar(50)	☒
	AFF_PHONE1	varchar(50)	☒
	AFF_EMAILID	varchar(100)	☒
	AFF_StatementStatus	varchar(50)	☒
	AFF_STATUS_TEXT	varchar(50)	☒
	IsAutoPayEnabled	bit	☒
	AFF_OpportunityID	varchar(50)	☒
	AgreementSigned	datetime	☒
	AccountManager	varchar(50)	☒
	AFF_CreatedBy	varchar(50)	☒
	AFF_ReasonLost	int	☒
	CreatedDateTime	datetime	☐
	AFF_CloseDate	datetime	☒
	DeactivatedAt	datetime	☒
	SFAAlertMessage	varchar(255)	☒
	AFF_TimeZone	varchar(50)	☒
	PrimaryUser	bigint	☒
	AFF_Latitude	float	☒
	AFF_Longitude	float	☒
	TowerRating	varchar(10)	☒
	DealerLicense	varchar(30)	☒
	IsNoTitleEnabled	bit	☐

6. IMPLEMENTATION AND PLANNING

6.1 IMPLEMENTATION ENVIRONMENT

Tools Used for Implementation:

- JetBrains Rider IDE 2023.3
- Dot Net 8
- RabbitMQ 3.9.0

Multi-User vs Single-User

Single user applications are the application where it is useful to only one user at a time. While in Multi user given application is used by many users at the same time and thus application is used by many users at the same time. Our Android Application is a multiuser system as we have more than one user who can use the system at the same time.

GUI vs non-GUI

Non-GUI application uses command Prompt for input and output while GUI application has graphics form to interface and other graphics property for various I/O operations and are easy to use Our System is a GUI based and thus easy and effective to use therefore user can easily give input and take Input.

Pre requisites:

- C#
- Angular
- Asp.Net Core
- REST API

6.2 PROGRAM/MODULE SPECIFICATION

- ZIP code Modules
- Select a car year Module
- Key and working battery Module
- Drive Module
- Body panel Module
- Instant Bid Module
- Campaign Module
- Assignment Module

6.3 CODING STANDARD

Coding Standards contribute to an improved comprehension of source code. Perhaps one of the most influential aids to understanding the logical flow of an application is how the various elements of the application are named. A name should tell "what" rather than "how". By avoiding names that expose the underlying implementation, which can change, you preserve a layer of abstraction that simplifies the complexity. Naming Conventions make programs more understandable by making them easier to read. They can also give information about the function of the identifier – for example, whether it's a constant, class, etc. which can be helpful in understanding the code.

All code should be well commented. All procedures and functions should begin with a comment to explain what the function/procedure performs.

- Good and meaningful comments make code more maintainable.
- Do not write comments for every line of code and every variable defined.

Write comments wherever required. But good readable code will require very less comments. If all the variables and methods names are meaningful, that would make the code very readable and will not need more comments.

Reasons for using the coding standards are:

- Uniform distribution
- Sound understanding
- Encourages Good programming skills.

7. TESTING

7.1 TESTING PLAN

The different types of testing methods used in the project are:

1. Unit Testing
2. UI Testing
3. Performance Testing
4. System Testing

Requirement of Testing

How can we find defects and errors in the code and design of every programmer? After testing, we make sure that our code produces desirable results, and if there are any flaws, we may fix them before releasing it.

7.2 TESTING STRATEGY

In the software development life cycle, after the requirement analysis, feasibility study, design and coding phase, one of the phases of utmost importance is the testing phase. In this phase, we get to see if the expected and final outcomes are the same with all the required specifications being maintained as per the requirement. This is the debugging phase of the application in its entirety and not simple debugging of a few lines of code on the IDE. This phase starts with the smallest unit of a code through Unit Testing and travels a long way till it ends with User Acceptance Testing. For industrial applications, testing is performed through highly automated tools like Docker for NodeJS applications, but such automated testing tools are out of scope for this project. This testing phase helps in finding bugs which can be at the code level, system level, environment level, then fixing those bugs, then retesting the same functionalities to check the new changes are compatible with others and then at the end of many testing strategies the product is delegated to corresponding authority for release. Testing at the application/code level is performed by writing test cases. To err is human – we all know. When code is nothing but human typed language with a sense of grammar and context understandable by the OS and computer scientists, errors can always exist. We can eliminate these errors by thinking about corner cases or scenarios that might never happen but if they happen, our application is strong enough to handle them. Testing the application with every possible scenario which it can/cannot handle and still our application staying steadfast is what we target for.

Testing objectives:

- Testing is a process of executing a program with the intention of finding an error.
- A good test case is one that has a high probability of finding an as-yet undiscovered error.
- A successful test is one that uncover an as-yet undiscovered error.

7.2.1 Unit Testing

Unit testing is a software development process in which the smallest testable part of an application, called units, are individually scrutinized for proper operation. Unit testing is often automated but it can also be done manually. This testing mode is a component of Extreme Programming (XP), a pragmatic method of software development that takes a meticulous approach to building a product by means of continual testing and revision.

Unit testing involves only those characteristics that are vital to the performance of the unit under test. This encourages developers to modify the source code without immediate concerns about how such changes might affect the functioning of the units or the program as a whole. Once all of the units in a program have been found to be working in the most efficient and error free manner possible, larger components of the program can be evaluated by means of integration testing. I tested each single part of the entire application. I tested each and every module individually.

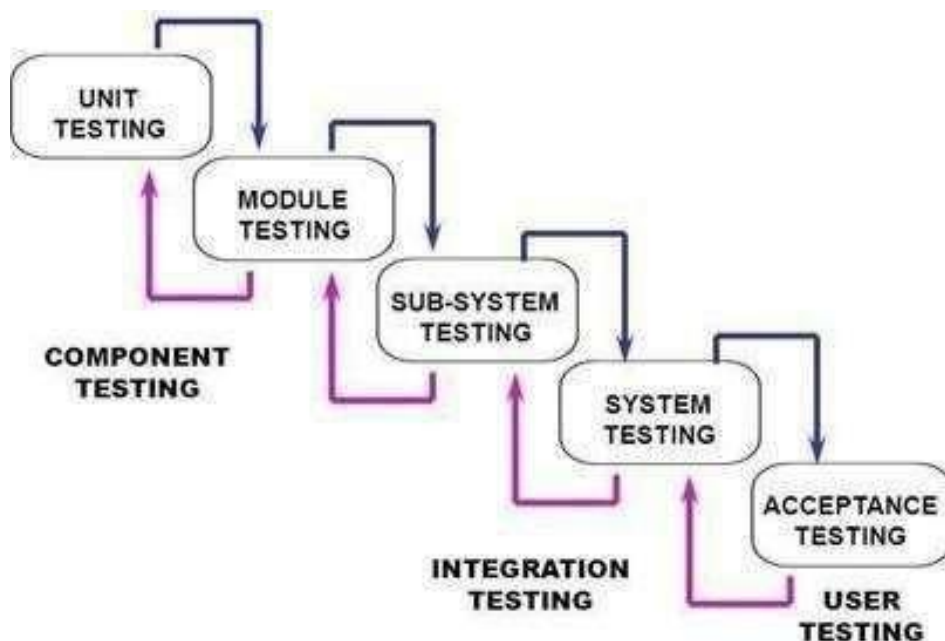


Fig. 7.1 Types of Testing

Ref. <https://images.app.goo.gl/BNdaJFa1vB4AYYQi6>

7.2.2 Sub System Testing

After testing each unit, we move on to larger units called sub systems. In subsystem testing we tested the whole user side as one system. On the user side all the modules like dashboard, API, etc. were tested together to see if there was any error or bug found.

7.2.3 System Testing

After testing all the sub-system, it is time to test the whole system. System testing of software is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. While testing the whole system we found many errors like the data of the applicants were not displayed on the dashboard of the Employer for a particular type of job category and also there was a problem in sending the invitation.

- **Recovery Testing:** It is a system test that forces the software to fail in a variety of ways and verifies that recovery is properly performed.
- **Security Testing:** It attempts to verify that protection mechanisms build into a system will, in fact, protect it from improper penetration.
- **Performance Testing:** It is designed to test the run-time performance of software within the context of an integrated system performance testing occurs throughout all steps in the testing process.

7.2.4 Acceptance Testing

Acceptance testing can be connected by the end user, customer, or client to validate whether or not to accept the product. Acceptance testing may be performed as part of the hand-off process between any two phases of development. The acceptance test suite is run against the supplied input data or using an acceptance test script to direct the tester. Then the results obtained are compared with the expected results. If there is a correct match for every case, the test suite is said to pass.

7.3 TESTING METHODS

The verification activities fall into the category of static testing. During static testing, you have a check list to check whether the work you are doing is going as per the set standards of the organization. These standards can be for coding, integrating and deployment. Reviews, Inspections and Walkthroughs are static testing methodology. Dynamic testing involves working with the software giving input values and checking if the output is as expected. These are the validation activities. Unit test, integration test, System and acceptance tests are few of the dynamic testing methodologies.

Alpha & beta testing: the alpha test is conducted at the developer's site by a customer. The software is used in a natural setting with the developer "looking over shoulder" of the user and recording errors and usage problems. Alpha tests are conducted in a controlled environment. The beta testing is conducted at one or more customer sites by the end-user of the software. Unlike alpha testing, the developer is generally not present. Therefore, the beta test is a "live" application of the software in an environment that cannot be controlled by the developer.

7.3.1 Black Box Testing

Also known as functional testing. Software testing techniques where the internal working of the item being tested are not known by the tester. For example, in a black box test on software design the tester only knows the inputs and what the expected

outcomes should be and not how the program arrives at those outputs. The tester does not ever examine the programming code and does not need any further knowledge of the program other than its specification.

The advantages of this type of testing include:

- The test is unbiased as the designer and the tester are independent of each other
- The tester does not need knowledge of any specific programming languages
- The test is done from the point of view of the user, not the designer
- Test cases can be designed as soon as the specifications are complete.

The disadvantages of this type of testing include:

- The test can be redundant if the software designer has already run a test case
- The test cases are difficult to design
- Testing every possible input stream is unrealistic because it would take an inordinate amount of time: hence many program paths will go untested.

7.3.2 White Box Testing

Also known as glass box, structural, clear box and open box testing. A software testing technique whereby explicit knowledge of the internal workings of the item being tested are used to select the test data. Unlike black box testing, white box testing uses specific knowledge of programming code to examine outputs. The test is accurate only if the tester knows what the program is supposed to do. He or she can then see if the program diverges from its intended goal.

7.4 TEST CASES

To minimize the number of errors in software, a rich variety of test design methods have evolved for software. These methods provide the developer with a systematic approach to testing. More importantly, methods provide a mechanism that can help to ensure the completeness of the test and provide the highest likelihood for uncovering errors in software. An engineering product can be tested in one of the two ways:

(1) knowing the specified function that product has been designed to perform, tests can be conducted that demonstrate each function is fully operational while at the same time searching for errors in each function:

(2) knowing the internal workings of a product, tests can be conducted to ensure that “all gear mesh”, that is, internal oppression are performed according to specifications and all internal components have been adequately exercised. Here are the test cases that we had made for our application.

Here are some of the test cases that we created:

Table 7.1 Test Cases

Description	Condition	Expected	Actual	Pass/Fail
Register User	Same Phone No.	Doesn't Allow	Doesn't Allow	✓
	Different Phone No.	Allow	Allow	✓
Signup User	Enter all details valid	User created	User created	✓
Assignment	Enter valid details with region	Assignment created	Assignment created	✓
	Enter invalid details with region	Assignment not created	Assignment not created	✓
Edit profile	Enter valid data	Profile updated	Profile updated	✓
	Enter invalid data	Profile not updated	Profile not updated	✓
Campaign	Enter valid region and Ids	Campaign created	Campaign created	✓
	Enter invalid region and Ids	Campaign not created	Campaign not created	✓
Instant Bid	Enter bid value	Bid Created	Bid created	✓

Payment	Enter valid card details	Payment success	Payment success	✓
	Enter invalid card details	Payment fails	Payment fails	✓
Notification	If user Id exists	Notification Aries	Notification Aries	✓
	If user Id not exists	Notification not arrived	Notification not arrived	✓
Invoices	If payment success and user is valid	Invoice should be generated	Invoice should be generated	✓

7.5 CSRF (CROSS-SITE REQUEST FORGERY) TOKEN

Attackers can send URLs for login, logout, payment etc. and if the victim clicks on that link, the victim performs the task. It is called cross site request forgery. For preventing this attack CSRF token is come into picture. So if the victim clicks on the attacker's link, then it gives invalid CSRF token because it is different for every request and does not create cookies and also cannot access the account. So, we use CSRF Token.

8. USER MANUAL

8.1 Home Page:

- Displays the landing page offering fast cash deals for cars with options to get an instant quote.



Fig. 8.1 Home page

8.2 Sign Up page:

- Shows the user login screen with email/password and social login options.

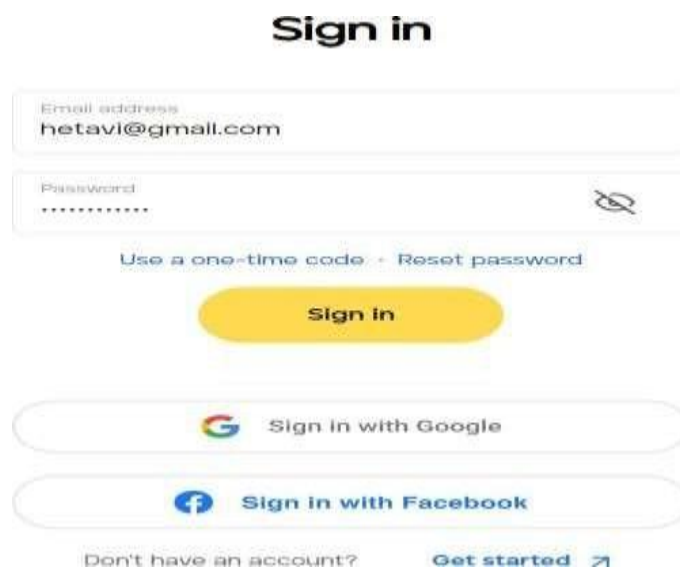


Fig. 8.2 Sign Up page

8.3 Steps to get car:

- Highlights the initial step of entering car details to create a campaign.

3 easy steps to get your car

Now it's easy to get your car

Tell us a little about your car.

AutoFill with VIN

Year	Make
Model	Trim (Optional)

STEPS TO GET ASSIGNMENT FOR CAR

CREATE CAMPAIGN
Create your campaign by adding your locations and car details.

CREATE INSTANT BID
Create your instant bids by adding your desired amount for particulate car.

3 easy steps to get your car

Now it's easy to get your car

Tell us a little about your car.

AutoFill with VIN

Year 2020	Make Audi
Model A3 Sedan	Model Audi A3 Audi A3 Sportback Audi A3 Sedan Audi A3 e-tron Audi A3 e-tron Sportback

Next

STEPS TO GET ASSIGNMENT FOR CAR

CREATE CAMPAIGN
Create your campaign by adding your locations and car details.

CREATE INSTANT BID
Create your instant bids by adding your desired amount for particulate car.

Fig. 8.3 Steps to get car

8.4 Steps to get car:

- Illustrates the mobile view of completing car assignment steps via campaign and instant bid.

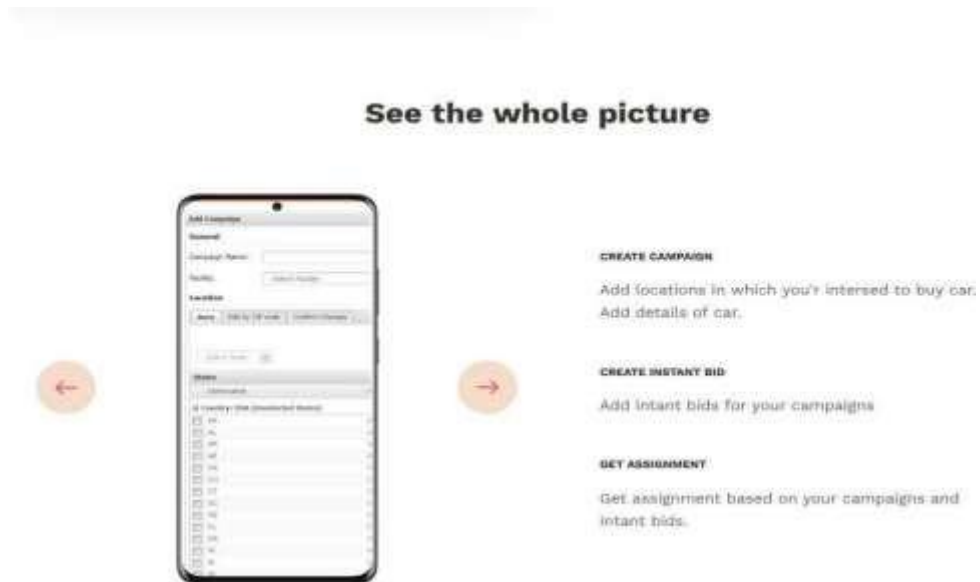


Fig. 8.4Steps to get car

8.5 Campaign Page:

- Displays a campaign management dashboard listing campaign names, bids, and update statuses.

Campaigns					
☐	Name ▲	Instant Bids	Facility	Conversion	Last Modified
☐	alaska jay	0	Do Not Update	0%	02/08/2023
☐	Copy campaign - 1	0	Do Not Update	0%	12/19/2022
☐	copy copy	0	Do Not Update	0%	02/08/2023
☐	Copy to do not update	0	Do Not Update	0%	12/19/2022
☐	jaajjja	0	Do Not Update	0%	03/09/2023
☐	only NY-10001	0	Do Not Update	0%	02/08/2023
☐	premier test campaign	0	Do Not Update	0%	02/08/2023
☐	test	0	Do Not Update	0%	12/18/2022
☐	Test campaign - 1	0	Do Not Update	0%	12/19/2022
☐	Test campaign - 2	0	Do Not Update	0%	12/19/2022
☐	Test copy	1	Do Not Update	0%	12/19/2022
☐	Test copy - 1	-1	Do Not Update	0%	12/19/2022
☐	test purpose	0	Do Not Update	0%	12/18/2022
☐	title	2	Do Not Update	0%	08/21/2022
☐	uniqueOne	0	Do Not Update	0%	02/08/2023

Fig. 8.5 Campaign Page

8.6 Instant Bid Page:

- Displays a list of instant bids across various campaigns with details like amount and conversion.

Instant Bids - All Campaigns back to list							
	Amount	Per	Name ▲	Campaign	Facility	Conversion	Last Mod...
☐	\$5,000	Vehicle	1-Copy1	title	Do Not Update	0%	12/18/2022
☐	\$2,400	Vehicle	Do not update 1	title	Do Not Update	0%	12/18/2022
☐	\$2,400	Vehicle	Instant bid	Test copy	Do Not Update	0%	12/19/2022
☐	\$2,400	Vehicle	Instant bid	Test co...	Do Not Update	0%	12/19/2022

Fig. 8.6 Instant Bid Page

8.7 Go to assignments:

- Shows the homepage with navigation to the “Assignments” section..



Fig. 8.7 Go to assignments

8.8 Open assignments:

- Lists all active assignments with their status and scheduling options.

Schedule Complete Cancel Un-Cancel More actions					
Open Assignments					
	Assignment	Assigne...	Facility	Location	Status
☐	2 1025687 2007 Ford Focus	9/10/18	Do Not Update	New York, NY	Scheduled for 3/11/23 [X]
☐	1 1025665 1991 Mercury Gr...	9/10/18	Do Not Update	New York, NY	To schedule
☐	1025659 2001 Pontiac Gr...	9/10/18	Do Not Update	New York, NY	To schedule

Fig. 8.8 Open assignments

8.9 Closed Assignments:

- Displays completed or canceled assignments along with their details.

Actions All Time From To Go						
Closed Assignments						
	Assignment	Assigned	Facility	Location	Status	Closed ▾
☐	1025652 2003 Oldsmobile A...	09/10/2018	Do Not Update	New York, NY	Cancelled	12/27/20...
☐	1025667 2000 Chevrolet Ma...	09/10/2018	Do Not Update	New York, NY	Completed	11/22/2022
☐	1025673 2007 Toyota Prius	09/10/2018	Do Not Update	New York, NY	Completed	09/20/20...
☐	1025664 2003 Mitsubishi L...	09/10/2018	Do Not Update	New York, NY	Completed	09/19/20...
☐	1025671 1999 Mitsubishi M...	09/10/2018	Do Not Update	New York, NY	Completed	09/14/20...
☐	1025657 2000 Buick LeSabre	09/10/2018	Do Not Update	New York, NY	Completed	08/10/20...
☐	1025655 2002 Ford Escape	09/10/2018	Do Not Update	New York, NY	Completed	08/10/20...
☐	1025654 1998 Lincoln Town...	09/10/2018	Do Not Update	New York, NY	Completed	08/10/20...
☐	1025656 1995 Oldsmobile 88	09/10/2018	Do Not Update	New York, NY	Completed	08/10/20...

Fig. 8.9 Closed Assignments

8.10 Invoices Page:

- Shows invoice records filtered by facility and time range.



Fig. 8.10 Invoices Page

8.11 Dashboard:

- Provides summary metrics including total bids, assignments, completions, and averages

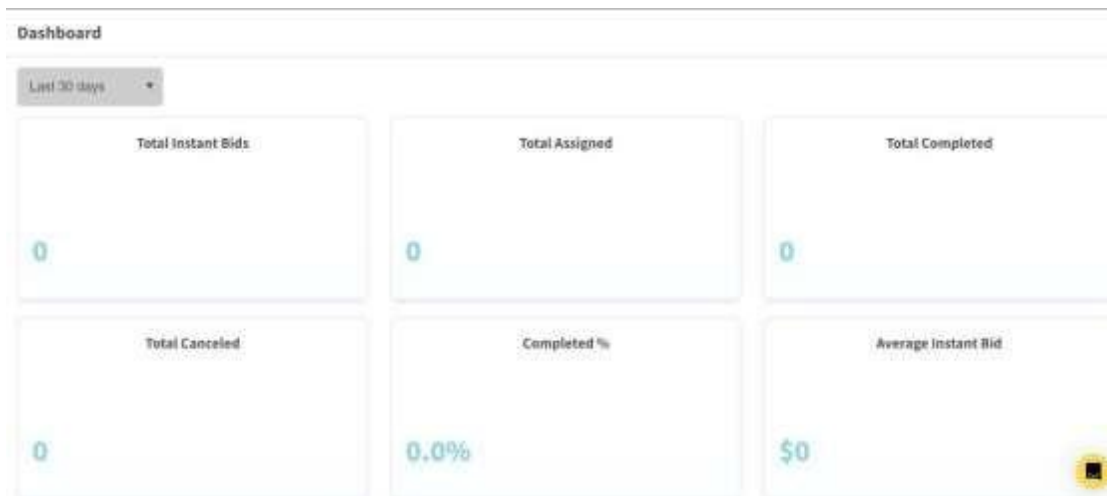


Fig. 8.11 Dashboard

8.12 Connect to us page:

- A simple footer page inviting users to create an account and connect with the company.

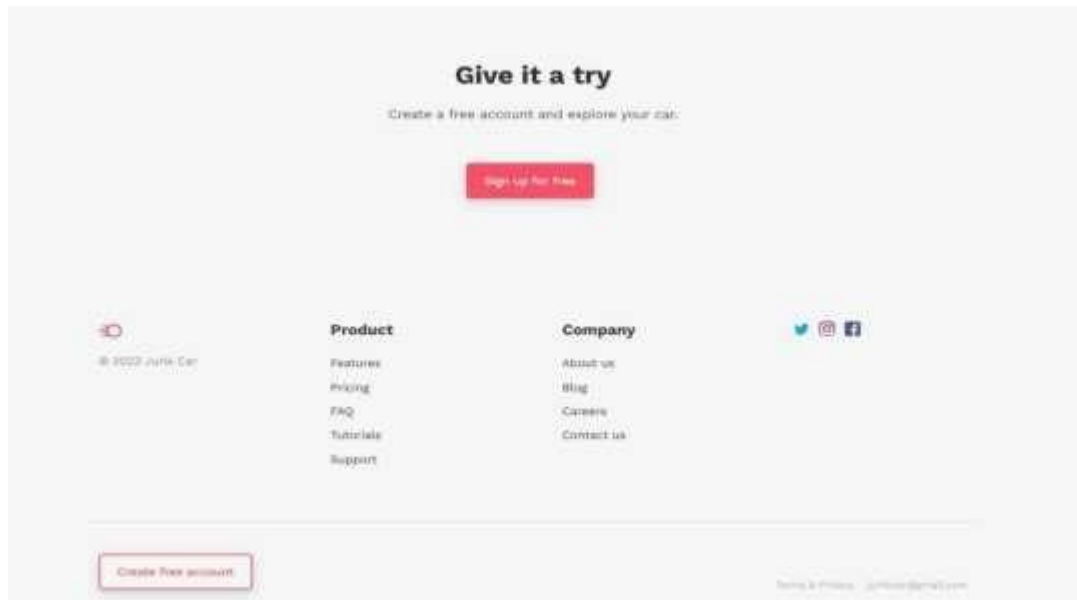


Fig. 8.12 Connect to us page

8.13 Assignment Showing Page:

- Displays a detailed offer with amount, pickup, and payment options for car assignment confirmation.

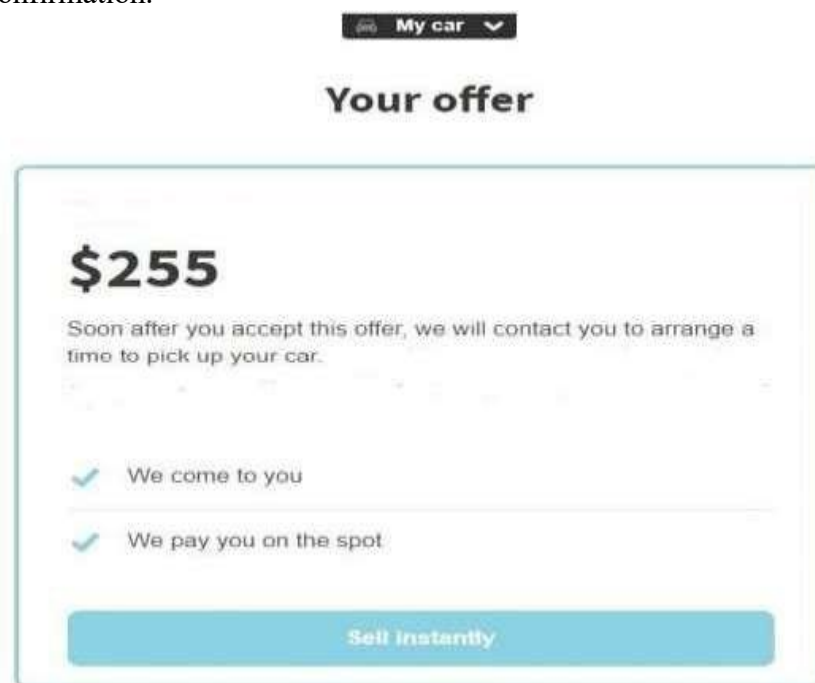


Fig. 8.13assignment showing page

9. LIMITATIONS AND FUTURE ENHANCEMENT

9.1 LIMITATIONS

- In current system value calculation of vehicle, we use zip to zip distance. So, we can't get the precise value of vehicle that satisfies both.
- If user enters incorrect details while creating instant bid or campaign it will calculate incorrect assignment. So, there is no validation from our side that check that entered details of campaign and instant bid are correct or not.
- In Current system Buyer can't Search Car based on Company Model, Year, City.

9.2 FUTURE ENHANCEMENT

- Well, I and my team members have worked hard in order to present an improved website better than the existing one's regarding the information about the various activities. Still, we found out that the project can be done in a better way.
- In assignment calculation we are using zip to zip distance to calculate the value of vehicle now in future we are planning to calculate the value on more precise distance.
- The next enhancement that we can add the notification system that notify buyer about available vehicle deals in he/her might be interested based on their pass assignments.
- The next enhancement will add is searching criteria for used cars will be more in variety that will be boost user interface.
- These are the enhancements that we could think of at present.

10. CONCLUSION AND DISCUSSION

10.1 CONCLUSION

- Car selling/buying business has emerged with a new goody compared to the past experience where every activity concerning car selling/buying a business is limited to a physical location only. It provides a friendly graphical user interface which proves to be better when compared to the existing system. It gives appropriate access to the authorized users depending on their permissions. Nowadays, customers can sell cars online, rent car online, and have the car brought to their door step once the customer is a registered member or go to the office to pick the car. The web-based car selling system has offered an advantage to both customers as well as Car buying Company to efficiently and effectively manage the business and satisfies customers' need at the click of a button.

10.2 DISCUSSION

- Overall, before developing the project while looking at the functionality – project seems to be very easy to implement. We thought that We'll just need to write the code now. But as We progressed in project, we faced many complex difficulties.
- Working with frontend – HTML, CSS was comparatively easy to learn than JavaScript. We thought as the design once ready can be reused for other activities as well, we just have to make some modifications that is required for the page.
- Working with .NET was first time for us and it was quite difficult but by help of my mentor and other senior colleagues of the company made I made it through and completed my work.

BIBLIOGRAPHY

- Microsoft .NET Documentation -<https://docs.microsoft.com/en-us/dotnet/core/getstarted>
- Seri log Documentation -<https://serilog.net/>
- New Relic Documentation -<https://docs.newrelic.com/>
- Plural Sight Learning portal -<https://www.pluralsight.com/>
- C# Documentation -<https://docs.microsoft.com/en-us/dotnet/csharp/>
- RabbitMQ Documentation -<https://www.rabbitmq.com/documentation.html>
- Health Check In ASP.NET Core -<https://docs.microsoft.com/en-us/aspnet/core/hostand-deploy/health-checks?view=aspnetcore-6.0>
- Visual Paradigm Software Diagram Design Tool -<https://www.visual-paradigm.com/>
- Slack API -<https://api.slack.com/>
- Post Man -<https://learning.postman.com/docs/getting-started/introduction/>